

AI Personal Assistant Project - My Understanding

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Project Overview

The project aims to build a Jarvis-style AI Personal Assistant that works across desktop and mobile devices. Users can interact through voice or chat, and the assistant performs approved actions such as managing calendars, searching files, sending messages, joining meetings, generating summaries, setting reminders, and automating daily workflows.

Key Features

- Voice and chat interaction
- Google Calendar integration
- Google Drive file search
- Meeting recording (with permission), transcription, summaries, and action items
- Task management and reminders
- Messaging workflows with user confirmation
- Cross-device synchronization
- Activity history and permission management

High-Level Architecture

User (Voice/Text) → FastAPI Backend → LangGraph AI Agent → LLM (GPT/Gemini) → Tool Calling (Google Calendar, Drive, Gmail, etc.) → Response to User.

Technologies I Identified

Python, FastAPI, LangChain, LangGraph, LLMs (OpenAI/Gemini), Google APIs, OAuth, Vector Database, PostgreSQL, Redis, LiveKit (Voice), Docker, Cloud Deployment.

My Learning Goals

- Understand AI Agent workflows.
- Learn FastAPI for backend APIs.
- Explore LangChain and LangGraph for agent orchestration.
- Integrate Google services using APIs.
- Build permission-based automation features.
- Understand deployment and scalable AI architecture.

My Contribution Plan

I will first understand the project architecture, APIs, and agent workflow. Then I will contribute by implementing backend APIs, integrating external tools, testing AI workflows, fixing bugs, and gradually building new automation features while learning from the team.